



International Council on Systems Engineering
A better world through a systems approach

Sustainability WG

incose.org

Agenda

- Why is it Important?
- Mission, Purpose & Outcomes
- Sustainability WG Activities
- How to Get Involved?
- Contact

What is Systems Engineering?

Systems Engineering is a **transdisciplinary** and **integrative** approach to enable the successful **realization**, **use**, and **retirement** of **engineered systems**, using systems principles and concepts, and scientific, technological, and management methods.

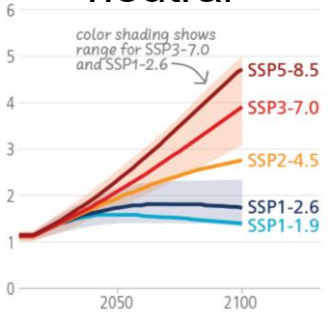
—International Council on Systems Engineering (INCOSE), 2022

Why is it important? (1/2): Sustainability

...many other planetary limits...

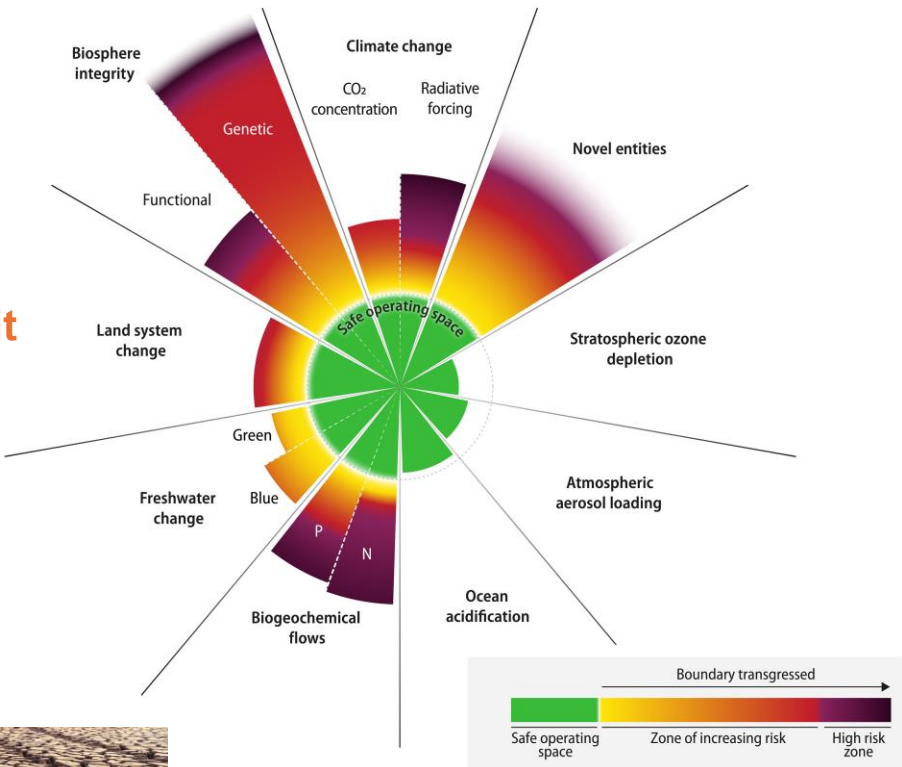
Climate change...

→ Accelerate toward carbon neutral

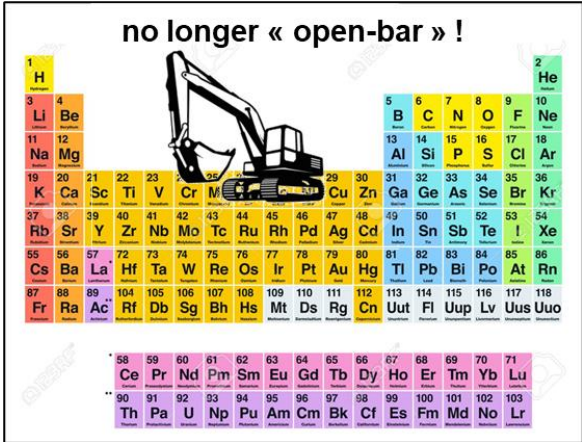


the COP21 1,5° limit has already been crossed!

→ Adapt to growing disruptions



... a Finite world...



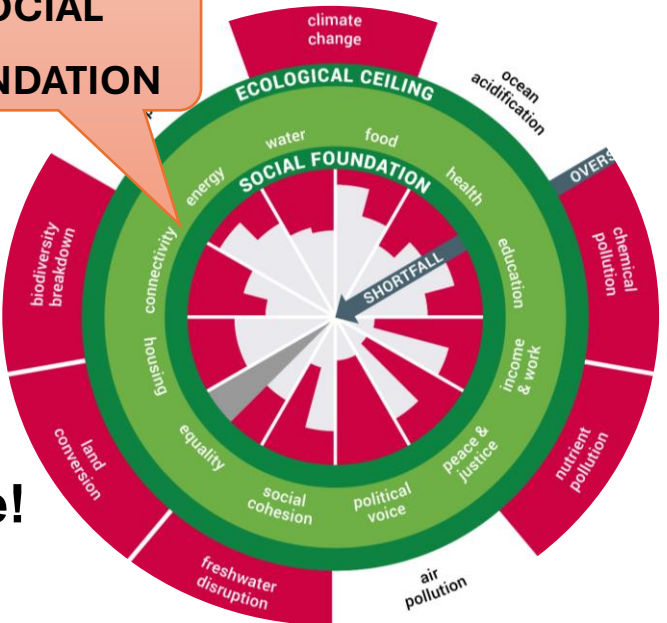
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...

Circular Economy

Cradle to Cradle

SOCIAL FOUNDATION



... with humans inside!

Why is it important? (2/2): a Sustainability Working Group @INCOSE

the first MEGATREND in INCOSE SE Vision 2035 !



Obviously, changes for technical systems...

- Stringent regulations, constraints,
- Evolving environmental context,
- New stakeholders,
- ...

... and new specialties, viewpoints, “models”, methodologies!

... but also for varied SOI and Environments ...

Socio-technical systems,
Services systems, PSS,
Enterprises,
Systems of Systems,
Eco-Systems,
Earth System!



Ecology Economy Environment
Politics Biology
Business model Biological Cycle
Technical Cycle
...

Mission, Vision & Outcomes

Mission

- The mission is to establish sustainability at the core of systems engineering enabled by systems thinking, considering all beings and elements globally. This will empower organizations to achieve sustainability in a finite world.

Vision

In line with the INCOSE SE Vision 2035 goals, we want to address the systems approach to sustainability:

- based on the vision of a finite world,
- where economic, social, and environmental aspects work in a synergistic and balanced manner,
- ...in order to meet the needs of the present without compromising those of future generations.

Initial Intended Outcomes

- INCOSE panel on “SOS and Sustainability” at SOSE conference in Tirana, Albania 2025 [\[complete\]](#)
- Formal public WG announcement at IS 2025 in Ottawa [\[complete\]](#)
- Sustainability Round Table at IW 2026, used to build awareness of the state of practice
- A new Knowledge Area in SEBoK curated by the WG introducing the topic, providing key references and reading lists
- Sustainability white paper/ Primer

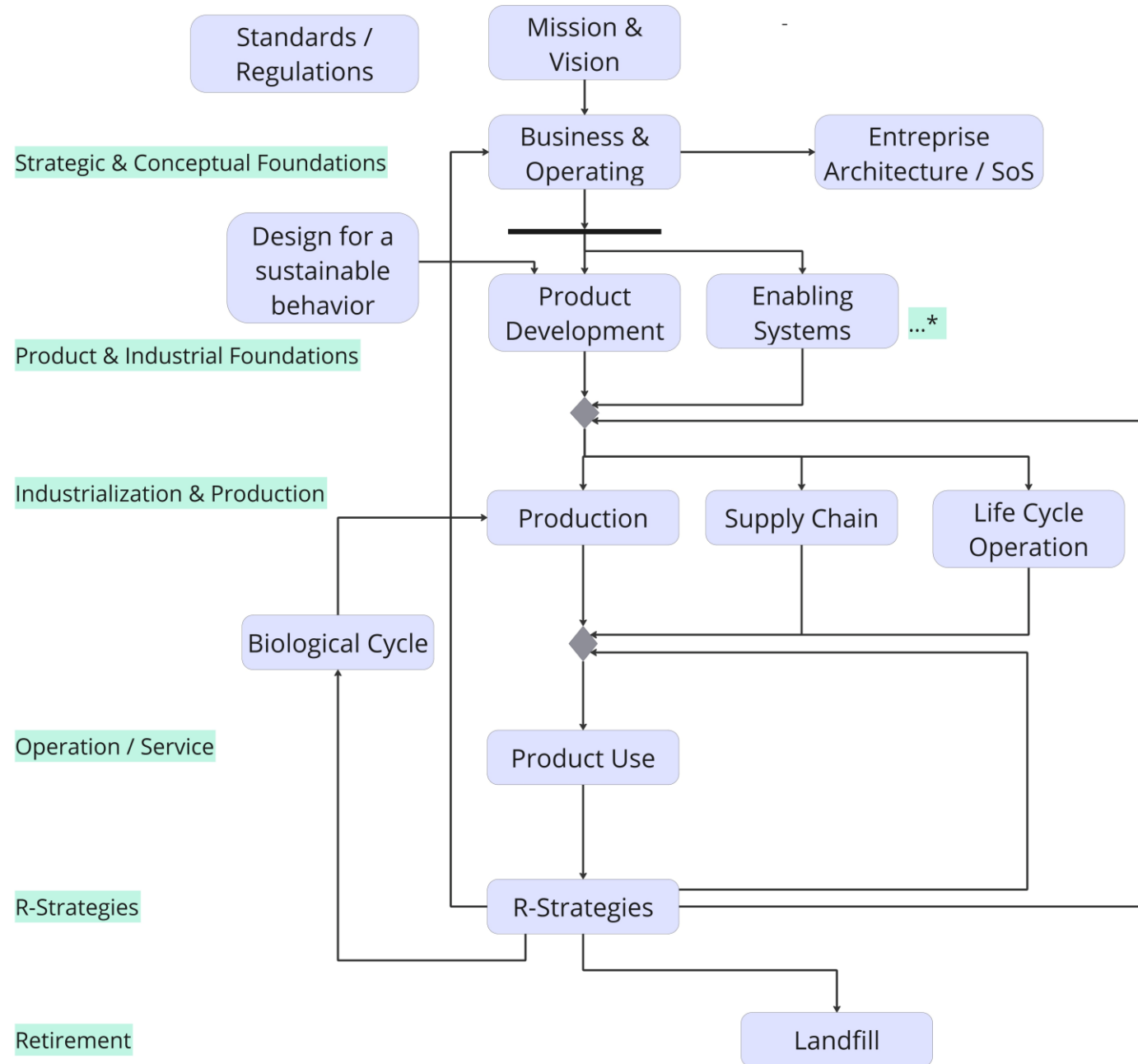
The INCOSE Sustainability WG was established in mid-February 2025.

Sustainability WG Activities

- SustainableTogether: *Transforming a conventional business model, enterprise and product into sustainable one. We show HOW to transform and demonstrate it using digital models.*
- CASCaRa Involvement: *A DPP Use Case implementation using CASCADE technology*
- Monthly Webinar Series on sustainability and systems engineering topics.
- Sustainability Primer / SEBoK article
- Outreach (Panel Discussion / Wildfire): *SoSE Panel Tirana 2025, Presentation at the IS2025, Project Wildfire paper being developed for IS 2026.*
- *Comparison Sustainability Standards/Frameworks*
- *Technosolutionism Risk*

Sustainable Together:

Initial Generic Approach of Sustainable-Together



- Systems Thinking and System Dynamics (Rebound-Effect) [Dano, Deepa]
- Digital Engineering [Fernando, Lambert, Gauthier, Deepa]
- Life Cycle assessment [Jean Francois, Lambert, Deepa]
- Risk Analysis [Hamza, Deepa]

SustainableTogether – Digital Engineering

CASCaDE Use Case

As part of the SustainableTogether project, we want to develop a knowledge graph to support the full product lifecycle. In parallel, the [CASCaDE](#) standard (Collaborative Artifact, Specification, Context, and Data Exchange) is being developed within OMG to enable standardized lifecycle collaboration through common data formats and ontologies.

By combining both initiatives, we are creating a joint use case to demonstrate their synergy. Current activities in SustainableTogether include:

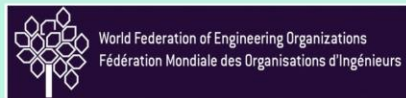
- Identifying digital engineering use cases
- Mapping required DPP data to the CASCaDE ontology

Who is involved in SustainableTogether?

SustainableTogether Network



Sustainability



WFEO



Education



Net-Positive
Advisory Panel



Flourishing
Business Model
Community



Circular
Economy



Cooperative
Business
Models

Systems Engineering



Sustainability WGs



Design Theory SIG



Monthly sustainability Webinars, 11 episodes and counting !

Upcoming episodes ...

Episodes 12&13

February 11th:

"Sustainable Systems Engineering: a practical model based on 'ecosystems thinking' to develop more regenerative systems."

Bert Van Lulling, Sies Overbeck

February 25th

An introduction to the Future-Fit Business Benchmark

Martin Rich

Episode 14 & 15

March 25th

Absolute Environmental Sustainability Assessment

Anders Bjoern, F. Paulilo AESA

March 30th

"Reboundless Design Theory"

Daniela Pigosso

Many
Episode Highlights , ie.
**Tim Lenton on Positive
Tipping Points in
December!**

Sustainability Webinar Series

Episode 1 June 2025
"Systemic Tools To Tackle Rebound Effects"
Ankita Das / Laetitia Borne

Episode 2 July 2025
"Systematically Designing Businesses Fit-for-the-Future"
Antony Upward

Episode 3 August 2025
"Defining Needs and Requirements for Sustainable Systems"
Lou Wheatcraft

Episode 4 September 2025
"From Net Zero to Net Positive: Driving Systemic Change Through Purposeful Business"
Franco Amalfi

Episode 5 : 29th of October 2025
"How Sustainability Standards Actually Undermine Sustainability and How to Hack Them to Achieve Authentic Sustainability"
Bill Baue

Episode 6: 12th of November 2025
"Collaborative Business Model for Circularity a Return on Experience and Puzzle based approach to Business Modelling"
Kalle Eldebo

Episode 7 : 26th of November 2025
Spaceship Earth From Attractive Metaphor to Architectural Analogy
Hillary Siletto

Sustainability Webinar Series

Episode 8 : 10th of December 2025
"Identifying Positive Tipping Points and Positive Tipping Point Framework"
Tim Lenton

Episode 9 : 17th of December 2025
"A Systemic perspective in circular business model patterns"
Christopher Dormeier

Episode 10: December 22nd 2025
Business Model LCA vs Product Level LCA: Introduction to BMCLA Method
Henrikke Baumann

Episode 11: 21st of January 2026
"Enterprise Sustainability Transformation Planning with UAF"
Aurelijus Morkevicius

Episode 12 & 13: February 11th
Sustainable Systems Engineering: a practical model based on 'ecosystems thinking' to develop more regenerative systems
Bert Van Lulling, Sies Overbeck

February 25th 2026
An introduction to the Future-Fit Business Benchmark
Martin Rich

Episode 14 & 15 March 25th 2026
Absolute Environmental Sustainability Assessment
Anders Bjoern, F. Paulilo AESA

March 30th 2026
Reboundless Design Theory
Daniela Pigosso

Sustainability Webinar Series

Episode 14 & 15 March 25th and 30th 2026

Absolute Environmental Sustainability Assessment AESA
Anders Bjoern, F. Paulilo

"Reboundless Design Theory"
Daniela Pigosso

Episode 16: April 8th 2026
"Systemic Perspective in process models for sustainability oriented business model innovation"
Christopher Dormeier

Episode 17: April 22nd 2026
Configuration Management & Sustainability
Sandrine Gonthier, Adriana De Souza ...

Episode 18: May 13th 2026
"Design theory for transition: Some insights from C-K design theory to manage creative preservation in systems engineering"
Pascal le Masson

Episode 19 : June TBD 2026
Milestone session accomplishments of Sust. WG so far

INCOSE Panel on Sustainable Development and Systems of Systems Engineering, Tirana 11th June 2025

Panellists from UK, Germany, France, Italy and Albania discussed:

- Influences of sustainable development on SoS engineering
- Examples of where SoS are being used to address sustainability challenges
- New and emerging approaches to SoS engineering?
- How to develop the next generation of engineers to address Sustainability and SoS?



How to Get Involved?

- **SustainableTogether Meeting, focused on SustainableTogether Project and open to non-incose members**
Every second Monday, 17:00 CET
- **INCOSE Sustainability WG Meeting, focus on INCOSE activities, e.g.: Primer and SEBoK**
Every second Wednesday, 17:00 CET
- **Join us:** Email sustainability@incose.net to receive your invitation

Important Links

- [Our Miro-Board](#): as a collaboration platform for meetings, brainstorming, etc.
- [Our GitHub Repo](#): The source of Truth of the SustainableTogether Project
- [Telegram Channel](#) for asynchronous communication and chatting
- Youtube Channel of the Sustainability Webinar Series (TBD.)

Navigating the Miro Board

- Open [Our Miro Board](#) – this is the SustainableTogether board.
- This is the main board, which links to topic-specific boards (see slide 9).
- To find topic boards, press **Ctrl + F** and search for “SustainableTogether Boards”.
- Content is organized chronologically from left to right. The most recent meeting notes are on the far right.

Contact Us



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For more information contact sustainability@incose.net
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